

Reflection

In this task, I created a simple Maven project from scratch and learned how to organize a Java project using the standard Maven directory structure. I initially had some difficulty setting up the project correctly because I manually created the folders and accidentally named the Maven configuration file `pox.xml` instead of `pom.xml`. This helped me understand that Maven depends on the `pom.xml` file to identify and manage a project.

I then created the required `src/main/java` directory structure and wrote a Java program to print a square. At first, the program printed stars throughout the square, but I modified the logic so that stars were printed only along the boundary, creating a hollow square. This helped me understand how nested loops and conditional statements can be used to control the output of a program.

I also learned how Maven builds a Java project using commands such as `mvn clean package`, and how the compiled `.class` files are generated inside the `target` directory. When Java initially could not find the main class, I traced the problem back to the incorrect project structure and fixed the source directory and package declaration.

Finally, I learned how to use Git from the terminal to prepare the project for GitHub. I checked the repository status, verified the remote repository, added the project files, and prepared them for the first commit. I also learned why a `.gitignore` file is important for excluding Maven-generated files such as the `target` directory.

Overall, this task gave me practical experience with Maven project structure, Java compilation, debugging file and package issues, and basic Git/GitHub workflow. The problems I encountered were useful because they made me understand not just the commands, but also why each part of the project needs to be structured correctly.